



CAEM: Confidence-Aware Episodic Memory with Self-Improvement

Md. Aksan Gony Alif | Maliha Binte Shamim | Md. Mahmudur Rahman
Department of Computer Science & Engineering, BRAC University

Headline result (vs. matched zero-shot baseline B1)

Training-panel architectural-contribution slice (CAEM C2 vs. B1):

$\Delta\text{EM} = +20.3\%$ rel. $\Delta\text{CHM} = -29.4\%$ rel.

Pooled training-panel EM lifts from 0.483 at B1 to 0.581 at CAEM C2; pooled training-panel CHM falls from 0.143 to 0.101. CAEM beats every one of the eight external baselines on both axes at both reference cycles (8/8 wins-against). The retrieval-utility classifier ablation lifts **CommonsenseQA** EM by +14.0 pp and **TruthfulQA** EM by +10.0 pp by re-routing misconception-bait queries away from the retrieval-augmented tier.

Problem: hallucination as a deployment-safety issue

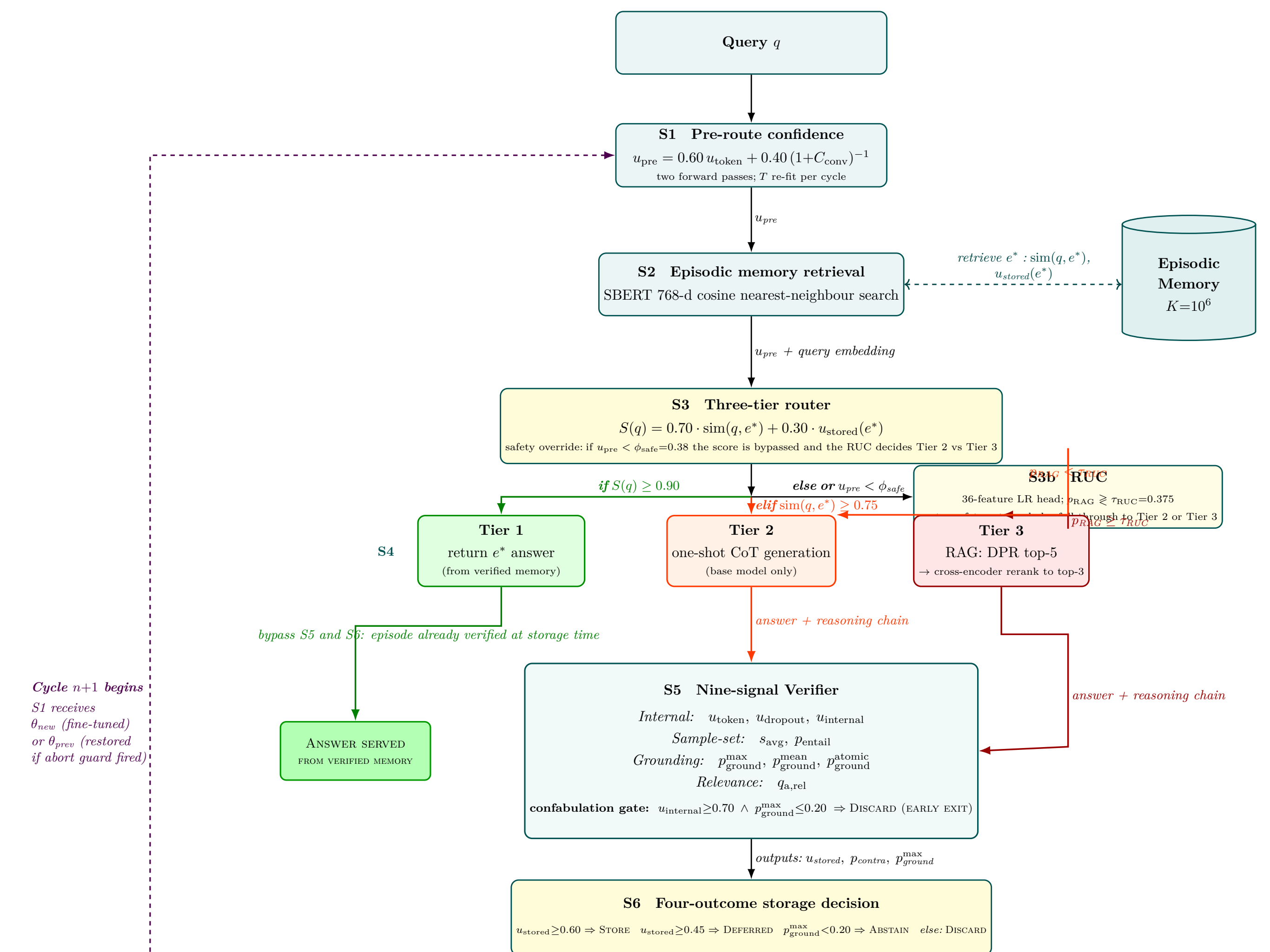
Modern instruction-tuned language models generate confident-sounding answers at deployment-relevant rates that exceed their actual accuracy by ten or more percentage points [1,2]. Single-signal correctness predictors achieve AUROC scores between 0.50 and 0.60, marginally above chance [3]. The TruthfulQA benchmark documents an inverse-scaling pattern under which larger models are *more* prone to echo human falsehoods [4]. Clinical-question studies report hallucination rates of 69 – 77% on frontier models [5].

The three structural shortcomings

- Statelessness:** no representation of past verified answers persists across sessions.
- Overconfidence:** single-signal correctness predictors are near random.
- Immutable capability:** naive online fine-tuning produces catastrophic forgetting.

CAEM is an architectural intervention that closes all three jointly.

The architecture: eight-stage pipeline + cycle-boundary loop



Nine-signal verifier + calibrated probability composite

Four signal families feed the per-benchmark calibrated probability composite:

- Internal calibration:** u_{token} , $u_{dropout}$, $u_{internal}$
- Sample-set agreement:** s_{avg} , p_{entail}
- External grounding:** p_{ground}^{max} , p_{ground}^{mean} , length-gated p_{ground}^{atomic}
- Question-answer relevance:** $q_{a,rel}$

Calibrated probability composite

Per-signal isotonic regression maps each raw signal to a calibrated probability; per-benchmark child curves are shrunk toward the pooled fit through a James-Stein-style prior at $\alpha = 0.6$; an L_2 -regularised logistic boost layer ($C = 0.01$) aggregates the per-signal log-odds into $u_{stored} \in [0, 1]$.

Storage gate (fixed threshold)

$$u_{stored} \geq 0.60 \rightarrow \text{Store} \quad u_{stored} \geq 0.45 \rightarrow \text{Defer} \quad \text{else} \rightarrow \text{Abstain / Discard}$$

Thresholds are frozen across cycles; only the composite refits per cycle. Replaces an earlier split-conformal gate rejected on the registered calibration-to-evaluation distribution-shift evidence.

Self-improvement loop with asymmetric-rollback retention guard

At every committed cycle boundary:

- Retroverify** every stored episode under the **locked verifier ensemble** (MiniCheck + frozen Qwen judge), re-scored against the updated generator's outputs and the cycle-boundary-refit composite. Prune entries that fall below $\tau_{retro} = 0.50$.
- Reconsider** every entry in the deferred buffer (promote, drop, keep) under $\text{TTL} = 2$ cycles.
- Consolidate** near-duplicate-meaning clusters under a cosine-similarity floor of 0.92.
- Fine-tune** the LoRA adapter on the strict training pool ($\tau_{train} = 0.70$); three epochs, effective batch 16.
- Probe** the registered multi-modal retention panel (MMLU, TQA-test, CSQA-test). If the worst-probe ratio drops below $p_{min} = 0.93$, the adapter rolls back to the previous committed checkpoint while **the memory store is preserved across the rollback** (the asymmetric-rollback contract).

Crucially, only the *generator* (advancing via SIL) and the *composite* (refitting per cycle) are cycle-indexed; the verifier ensemble is held fixed by design, so the per-cycle cal-fold class separation is a joint observable of the SIL-advanced generator viewed through that fixed lens.

Cycle-boundary trajectory: SIL + memory + commit pattern

Cycle	SIL n	Final loss	Worst probe	Commit	Memory pool
C1	107	0.258	1.006	✓	2 818
C2 ★	1 474	0.171	0.949	✓	5 151
C3	2 966	0.109	0.976	✓	7 477
C4	3 624	0.070	0.886 × ABORT		9 961
C5	5 006	0.091	0.958	✓	12 175

The realised commit pattern (1, 1, 1, 0, 1) yields commit fraction $\hat{\pi}_{commit} = 4/5 = 0.80$, strictly in the open interval (0, 1). The memory pool grows monotonically from 431 cold-start entries to 12175 at the cycle-five close **even across the cycle-four retention abort**, because the stream-chunk path executes independently of the retention guard under the asymmetric-rollback contract: deployment-time cost amortisation (memory tier growth) decouples from training-time parametric progression (adapter commits). Final-cycle losses decline monotonically across the four uninterrupted fine-tunes (0.258 $\rightarrow$ 0.171 $\rightarrow$ 0.109 $\rightarrow$ 0.070), the empirical receipt that SIL makes coherent gradient progress on the verified pool.

Per-cycle pooled trajectory (5-benchmark evaluation panel)

Cycle	Pooled EM	Pooled CHM	MMLU	TQA-test	CSQA-test
B1 (zero-shot ref)	0.507	0.120	—	—	—
C0	0.541	0.135	—	—	—
C1	0.543	0.112	1.041	1.013	1.006
C2 ★	0.544	0.107	0.975	0.949	0.982
C3	0.532	0.115	0.992	0.987	0.976
C4 abort	0.532	0.115	1.018	0.886	0.982
C5 (final)	0.517	0.109	0.999	1.013	0.958

The composite hallucination metric reduces –19% relative across the trajectory and attains its minimum at C2. C4 fires the retention guard on TQA-test = 0.886 < $p_{min} = 0.93$; C5 recovers to 0.958. Pool purity holds in the band [0.732, 0.746] on the cal fold across all committed cycles, clearing the registered 0.64 floor by 9 – 11 pp.

External eight-baseline panel (matched protocol)

System	EM	CHM	ΔEM vs C2	ΔCHM vs C2
★ CAEM C2 (best)	0.544	0.107	—	—
CAEM C5 (final)	0.517	0.109	−0.027	+0.002
B1 zero-shot	0.507	0.120	+7.4%	−11.0%
B2 chain-of-thought	0.500	0.123	+8.8%	−12.7%
B3 DPR-RAG	0.458	0.148	+18.8%	−27.7%
B4 CoT + DPR-RAG	0.461	0.142	+17.9%	−24.6%
B5 five-shot CoT	0.405	0.136	+34.2%	−21.4%
B6 vanilla fine-tune (C5)	0.325	0.144	+67.2%	−25.5%
B7 FLARE	0.292	0.153	+86.3%	−29.9%
B8 semantic entropy	0.505	0.120	+7.7%	−10.5%

Wins-against: CAEM C2 beats 8/8 baselines; CAEM C5 beats 8/8. CHM is post-hoc rescored through the locked CAEM verifier for every row, so the column is apples-to-apples. TruthfulQA EM uses the Haiku-judged correctness rule for every row including the abstention-licensed B8.

Retrieval-utility-classifier ablation (C3 close)

Benchmark	EM off	EM on	ΔEM (pp)	T3 share off	T3 share on	Δ
FEVER	0.503	0.503	0.0	92.3%	70.7%	−21.7
TriviaQA	0.437	0.463	+2.7	95.7%	68.0%	−27.7
CommonsenseQA	0.603	0.743	+14.0	57.7%	20.3%	−37.3
StrategyQA	0.617	0.640	+2.3	100.0%	76.7%	−23.3
TruthfulQA	0.210	0.310	+10.0	99.0%	40.0%	−59.0
Pooled	0.474	0.532	+5.8	88.9%	55.1%	−33.8

The asymmetric-rescue pattern (bold cells): the largest EM gains land on CommonsenseQA and TruthfulQA, the two benchmarks where retrieval was actively hurting in the classifier-off arm. FEVER EM stays flat: the classifier correctly preserves the retrieval-helpful regime on a textbook fact-verification benchmark.

Code, data, and references

Code: github.com/aksaN000/caem-thesis Artefacts (Google Drive): drive.google.com/drive/folders/1xErhsRDY_e_q00ZgUBn0WXXK81eHc5jZCa Passage index (HF, private): [aksaN000/caem-passage-index-21m](https://huggingface.co/aksaN000/caem-passage-index-21m)

[1] Guo et al., ICML 2017. [2] Kadavath et al., arXiv 2022. [3] Xiong et al., ICLR 2024. [4] Lin et al., ACL 2022. [5] Alkaissi & McFarlane, Curreus 2023. [6] Thorne et al., NAACL 2018. [7] Joshi et al., ACL 2017. [8] Talmor et al., NAAACL 2019. [9] Geva et al., TACL 2021. [10] Hu et al., ICLR 2022 (LoRA). [11] Tang et al., 2024 (MiniCheck). [12] Cherian et al., 2024 (boost layer).